

CANDIDATE'S NAME.....

SIGNATURE.....

Personal number



Apex Examination Council

PRE MOCK EXAMUNATIONS 2026

UGANDA ADVANCED CERTIFICATE OF EDUCATION

CHEMISTRY

PAPER 1

INSTRUCTIONS TO CANDIDATES

This paper consists two sections A and B. It has 6 examination items.

Section A has two compulsory items.

Section B has two parts. Part I and part II. Answer one item from each part.

Answer four items in all.

Any additional item(s) answered will not be scored.

SECTION A

Answer all questions in this section.

ITEM 1

SCENARIO

PhosGuard Industries manufacture phosphorus-derived reagents used in water treatment to remove heavy metal poisoning. Part of their continuous firing reactor line handles phosphorus(V) chloride (PCl_5) as an intermediate. The safety team performs gas phase equilibrium checks to ensure plant safety and downstream water treatment compatibility.

A gas phase equilibrium test was carried out by introducing 3.60 moles of phosphorus(V) chloride into a rigid vessel of volume 2.00 dm^3 at a given temperature. Analysis showed 1% dissociation according to:

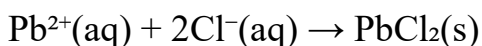
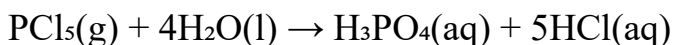


$$\Delta H = +124 \text{ kJ mol}^{-1}$$

Using this information, the equilibrium constant, K_c , can be determined.

In a follow-up process optimization test, the company introduces an additional 0.050 moles of chlorine gas (Cl_2) into the same system at the same temperature.

A treatment test was carried out where 0.208 g of phosphorus(V) chloride (PCl_5) was completely hydrolysed in 1.00 dm^3 of water, producing hydrochloric acid (HCl). The acid produced supplies chloride ions (Cl^-), which are then used to react with lead(II) ions (Pb^{2+}) in wastewater to form a precipitate.



A sample of wastewater contains $0.020 \text{ mol dm}^{-3}$ Pb^{2+} ions. You have been contacted for help.

TASK

As a learner of chemistry, guide the company on how to:

- Calculate the equilibrium constant K_c and comment on the extent of the reaction.
- Determine the new equilibrium concentrations of all species during the optimization test, hence determine the direction in which the equilibrium shifts and explain your answer.
- Explain what would happen to the equilibrium position, the equilibrium constant (K_c) and the speed of attainment of equilibrium when:
 - Temperature is increased.

- (ii) Pressure is decreased.
- (iii) Helium gas is added at constant volume.
- (d) Advise the company on how to optimize operating conditions (temperature, pressure and concentration) in order to control equilibrium effectively, improve production efficiency and ensure safety in handling phosphorus(V) chloride.
- (e) Calculate the maximum number of moles of Pb^{2+} ions that can be precipitated by the chloride ions present and comment on whether chloride ions are sufficient to completely remove Pb^{2+} ions from the wastewater.
- (f) Suggest possible environmental impacts that PhosGuard Industries should beware of and propose appropriate mitigation measures.

ITEM 2

SCENARIO

A chemical processing plant plans to scale up production of propan-2-ol, an important solvent used in sanitizers. Plant managers are evaluating two possible preparation routes, and management wants a full thermochemical assessment before commissioning either pathway.

The major preparation route involves hydrogenation of propanone.

Alternatively, propan-2-ol can be obtained from hydration of propene.

The thermochemical department has supplied bond energy data and auxiliary standard enthalpy information for use in the assessment.

To inform decision-making on the best preparation route, the manager has been advised that the most feasible process should be selected based on thermochemical considerations. You have therefore been contacted for assistance.

TASK

As a learner of chemistry, help the manager to:

- (a) Construct a Hess's law energy cycle for each preparation route using the thermochemical data provided.
- (b) Calculate the enthalpy change for:
 - (i) Hydrogenation of propanone.
 - (ii) Hydration of propene.
- (c) Evaluate which preparation route is more feasible for large-scale production of propan-2-ol.
- (d) Predict any major environmental impact arising from the process and propose a realistic mitigation measure.

SECTION B

PART I

ITEM 3

SCENARIO

Your chemistry class has been invited by a chemical plant management near Lake Victoria to assist in identifying ways of optimizing the production of key industrial compounds including sodium oxide, magnesium oxide and silicon dioxide used in glassmaking and ceramics.

The plant management has observed inconsistencies in melting points and reactivity, which are affecting product quality and safety. You are tasked with conducting a scientific investigation to analyse periodic trends of the elements, compound properties and molecular structures in order to recommend improvements.

Important chemical data concerning the elements and their compounds has been provided to assist in the analysis.

TASK

As a chemistry learner, write a report about the periodic trends of the elements and their oxides, and recommend improvements to the plant's production processes.

ITEM 4

SCENARIO

Uganda faces persistent challenges in rural electrification. Many communities rely on unreliable and costly energy sources such as diesel generators, kerosene lamps and car batteries.

A Ugandan company is developing solar-powered micro-grid systems and has invited your chemistry class to help identify suitable materials for solar batteries and wiring. These materials must be affordable, corrosion-resistant and durable under Uganda's rural conditions.

The company is investigating elements similar to those found in Periods 2 and 3 of the Periodic Table. Information on atomic radius, ionization energy, electronegativity and bonding type has been provided.

TASK

As a chemistry learner, use your understanding of atomic structure, periodic trends, chemical bonding and the information provided to propose suitable materials for solar batteries and wiring in Uganda's rural micro-grid systems.

PART II

ITEM 5

SCENARIO

A certain town in Central Uganda faces growing concerns over water pollution from agro-processing industries, especially during the cocoa harvest season. Wastewater from cocoa processing plants often contains organic matter, including acids and amines, which can harm aquatic life if released untreated.

A group of entrepreneurs is setting up a small plant to process cocoa husks, a by-product of cocoa bean production, into value-added products. Chemical analysis of the husks shows that they contain ethanol (from natural fermentation), ethanoic acid, small amounts of amines (from protein breakdown) and aromatic aldehydes such as vanillin (4-hydroxy-3-methoxybenzaldehyde).

To address the water pollution challenge and create additional revenue streams, the team plans to:

1. Determine the relationship between the structure of vanillin and ethanoic acid and their solubility.
2. Convert ethanol into amines for use in water treatment and cosmetics.
3. React ethanoic acid with ethanol to produce an ester for fragrances and study its reaction mechanism.
4. Evaluate the use of esters and amines in producing fragrances and cosmetics.

You have been tasked by your teacher to design feasible chemical processes, explain the underlying organic chemistry and propose sustainable solutions that will help the plant reduce environmental harm while maintaining profitability.

TASK

Design feasible chemical processes, explain the underlying organic chemistry and propose sustainable solutions that will help the plant reduce environmental harm while maintaining profitability.

ITEM 6

SCENARIO

In southwestern Uganda, a small cosmetics start-up is producing herbal skin-care creams using locally sourced plant oils such as shea butter and sunflower oil.

The production process generates significant quantities of waste plant oils and fats. These wastes are currently disposed of into nearby drainage systems, causing blockages and foul smells, which has led to complaints from the community and environmental authorities.

The company has approached a team of A-level chemistry students to:

- Analyse the composition of the waste materials, their functional groups and physical properties.
- Explore possible chemical processes that can convert them into valuable products and explain their mechanisms.
- Design a synthetic route to convert Compound C into Compound D.
- Propose a sustainable chemical process to convert Waste A into a useful marketable product.

Analysis of the waste oil revealed the following compounds:

A: $\text{CH}_3(\text{CH}_2)_{14}\text{COOCH}_2\text{CH}(\text{OH})\text{CH}_2\text{OH}$

B: $\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$

C: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

D: CH_3COCH_3

TASK

As a chemistry learner, give a written presentation that addresses the company's challenges and outlines scientifically sound and sustainable solutions.

END